

Holding a speech is Hard, Math is Harder? Hormonal and Physiological Stress Responses to the Speech-based versus Math-Based Trier Social Stress Test

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Abstract

Chronic exposure to stress is associated with adverse long-term consequences. However, the complex relationship between stressors, endocrine processes and resulting physiological and psychological outcomes remains insufficiently understood. This quasi-experimental study, conducted at the end of 2025 as part of the Digital Health Psychology seminar at Friedrich-Alexander University Erlangen-Nuremberg, used the Trier Social Stress Test (TSST) paradigm to examine stress responses under two conditions: a mental arithmetic task and a speech task. The primary outcome was the induced stress response, assessed using a multimodal approach, including physiological measures (heart rate, heart rate variability, cortisol- and alpha-amylase levels), behavioral indicators (body movement) and self-reported affect using the Positive and Negative Affect Schedule (PANAS). It was hypothesized that cortisol responses, autonomic stress responses and affective changes from pre- to post-TSST would differ between the conditions. The convenience sample consisted of 11 participants. The results showed no significant differences between the two conditions in cortisol levels, autonomic stress responses (i.e. alpha-amylase levels, HR and HRV) and positive and negative affect. However, a trend indicated a greater increase in peak cortisol levels in the mental arithmetic condition. These findings should be interpreted with caution, as the small sample size limited statistical power, so potential effects may therefore not have been detected. Future studies with larger sample sizes and a better control of confounding variables are needed to further investigate these trends and to clarify potential differences in stress responses under various TSST conditions.

Keywords: trier social stress test, stress response, cortisol response, autonomic stress response

- Marina Cörlin

Introduction

Even though stress is seen as a universal human experience, chronic exposure can lead to negative long-term impacts. While acute stress functions as an adaption to environmental demands, continuous stress response activation can become maladaptive (Schneiderman et al., 2005): the long-term effects of prolonged stress exposure can increase the risk for diseases like mood disorders, cardiovascular diseases, inflammation, autoimmune diseases and immune dysfunction, digestive issues and heightened morbidity, especially for a more vulnerable population (Montgomery et al., 2024; Schneiderman et al., 2005). Changes in the environment lead to a physiological adaptation, so-called allostasis, to ensure that vital systems are kept in the necessary balance needed for survival (Stephens & Wand, 2012) - however, prolonged exposure to a stressor can lead to overuse of the adaptive working systems, a state of allostatic overload, that can damage the body over the long term (McEwen, 2017). While responses to stressors have been researched on multiple levels, the full relationship between stressors, activated endocrine processes, and physiological, psychological, and cognitive outcomes is not yet fully understood. To advance physiological understanding as well as preventative care, understanding the working pathways of human stress response systems and interindividual differences in responses to various stressors is of central interest.

Physiological Stress Response Systems

When individuals experience stress, cognitive and affective responses have been shown to activate two main biological stress systems: The autonomic nervous system (ANS) and the Hypothalamic-Pituitary-Adrenal (HPA) axis (Ringgold et al., 2025). Activation of the HPA axis leads to a cascade of endocrine activity: The paraventricular nucleus (PVN) releases the neurohormones CRF and arginine vasopressin (AVP), leading to the stimulation of the anterior pituitary gland to produce adrenocorticotrophic hormone (ACTH). The secretion of ACTH activates the adrenal glands, which release cortisol (Stephens & Wand, 2012). Excitatory and inhibitory feedback loops work to maintain homeostasis and control for overly prolonged activity. In the body, the release of cortisol leads to an increase in blood glucose levels and other adaptations of the metabolism, but it also increases arousal, blood pressure, and anti-inflammatory immunological responses (Stephens & Wand, 2012) – all in the name of optimal physiological functioning during acute stressful and possibly dangerous situations.

Activation of the other main biological stress response system, the ANS, leads to immediate, short-lived reactivity of the sympathetic and parasympathetic nervous system, inducing the so-called “fight-or-flight” response (Ulrich-Lai & Herman, 2009). Especially the activation of the sympathetic-adrenal-medullary (SAM) system leads to secretion of epinephrine and norepinephrine, causing immediate effects on alertness, heart rate, and metabolic activity (Godoy et al., 2018). Both

the HPA axis and the SAM system interact in their activity to cause increased reactivity and adaptation to stress.

Measuring Stress Responses

Reactivity to stress can be measured by capturing the activity of the previously explained physiological stress systems. Activity of the HPA axis can be quantified through the measurement of cortisol concentration in the body – blood, saliva, urine, and hair are viable media to measure cortisol concentration. Salivary measures of cortisol concentration offer a non-invasive, economic, and accurate method to capture current HPA axis activity as a measure of stress reactivity (Smyth & Clow, 2020). Additionally, salivary alpha-amylase (sAA) levels have been shown to increase with SAM system excitation, indicating activity of the sympathetic and parasympathetic nervous system (Nater & Rohleder, 2009), therefore posing another biomarker for the measurement of physiological activity in response to environmental stressors. Finally, research has identified non-invasively observable biomarkers of stress responses such as heart rate (HR), heart rate variability (HRV) and body movements: While an HR increase is an indicator of ANS activity (Ringgold et al., 2025), HRV was shown to be impacted by stress, specifically through low activity of the parasympathetic nervous system (Kim et al., 2018). Lastly, Abel and colleagues (2025) showed the interrelation of body posture and movement patterns with physiological stress responses by finding a correlation between movement data during laboratory stress tests and cortisol reactivity.

- Lina Hävecker

Moderators of Stress Responses

A growing body of research suggests that cortisol responses are influenced by several moderators and thus show interindividual differences in cortisol reactivity. In this context, not only sociodemographic factors, such as biological sex and chronological age are relevant, but also psychopathological processes, anticipatory cognitive processes and task-related factors such as language use.

Regarding sociodemographic moderators of cortisol stress response, biological sex and chronological age represent robust, interrelated examples. While Larsson et al., 2009 displayed a higher basal cortisol concentrations in chronologically older individuals, experimental studies additionally demonstrate age-dependent differences in the magnitude and temporal dynamics of cortisol responses, highlighting aging-related modulation of stress physiology using the TSST (Kudielka et al., 2004). Similarly, sex differences were observed in several studies, indicating greater cortisol responses in men than in women (Reschke-Hernández et al., 2017; Liu et al., 2017; Gu et al., 2022). For example, Reschke-Hernández et al. (2017) demonstrated these differences in acute stress

reactivity when comparing two alternative tests (TSST and Iowa Singing Social Stress Test), despite comparable levels of subjective stress. These findings suggest sex-specific patterns of HPA axis activation rather than differences in perceived stress alone, however question has to be raised about the great variability of sampling procedures, which might mitigate the heterogeneity between sexes. Furthermore, gender differences may depend on the use of oral contraceptives, which primarily affect female individuals and cause mitigated cortisol response (e.g., Gervasio et al., 2022). Considering age and gender concurrently, it has been observed that cortisol secretion decreases with age in men and increases in women (e.g., Keenan et al., 2009), which raises important implications in terms of conjointly investigation of sociodemographic indicators (e.g., Halbreich et al., 1984).

Additionally, research shows differences in stress response in the context of psychopathological processes. While meta-analytical evidence does not provide clear evidence of differences in basal cortisol reactivity between depressed and control persons, indicating a relatively preserved cortisol reactivity in individuals with major depressive disorder (e.g., Knorr et al., 2010), studies focusing on the TSST show that individuals with depressive symptoms exhibited steeper reactivity and recovery to acute stressors (Chopra et al., 2009, Fiksdal et al., 2019). Regarding further psychopathological processes, meta-analytic evidence indicates that individuals with schizophrenia show attenuated HPA axis responses to social stressors compared to healthy controls (Ciufolini et al., 2014). In anxiety-related disorders, such as social phobia, subjective stress experiences may diverge from physiological stress markers. Klumbies et al. (2014) reported pronounced subjective stress responses in socially phobic individuals during a social stress task, accompanied by blunted or absent cortisol increases, underscoring a dissociation between experiential and endocrine stress components.

Beyond demographic and psychopathological factors, psychological and cognitive factors preceding stress exposure also play a key moderating role. Gaab et al. (2005) demonstrated that anticipatory cognitive appraisal prior to stress exposure significantly predicts cortisol responses to the TSST. Individuals who appraised the upcoming task as more threatening or uncontrollable showed increased cortisol responses, independent of the objective task demands.

Finally, emerging research highlights task-related contextual moderators, such as language use. Hauck et al. (2022) showed that performing an acute social stress task in a second language elicited stronger cortisol responses compared to performing the task in one's mother tongue. This suggests that increased cognitive load, linguistic insecurity, or heightened self-monitoring in a non-native language may amplify physiological stress responses. Such findings are particularly relevant for multilingual samples and cross-cultural stress research.

Taken together, these findings demonstrate that cortisol responses to social stressors are shaped by a complex interplay of sociodemographic and health-related indicators, as well as

anticipatory cognitive processes and contextual task characteristics. Accounting for the multifactorial nature of the moderators is essential for the accurate interpretation of HPA axis outcomes in experimental stress research.

- Ann-Kristin Reinhard

The Trier Social Stress Test (TSST)

The Trier Social Stress Test (TSST) is one of the most known and widely used, well-validated experimental paradigms for inducing acute psychosocial stress in humans. Since its introduction by Kirschbaum et al. (1993), the TSST has been employed extensively to investigate psychobiological stress responses, particularly activation of the hypothalamic–pituitary–adrenal (HPA) axis and the sympathetic-adreno-medullary (SAM) system. The TSST reliably combines two core stress-inducing elements: social-evaluative threat and uncontrollability, which together are known to elicit robust endocrine, physiological, cognitive, and affective stress responses (Labuschagne et al., 2019).

In the standard TSST protocol, participants first undergo a brief preparation period before completing two stress-inducing tasks in the presence of an evaluating panel. These tasks typically include a simulated job interview, followed by a mental arithmetic challenge. Participants are made aware that their performance will be observed and recorded, which contributes to the experience of social-evaluative threat. The arithmetic task further increases stress by imposing high cognitive demands and limited control, as individuals must perform serial calculations under time pressure while receiving immediate correction for errors (Labuschagne et al., 2019).

As Man et al. (2023) showed in their meta-analysis, salivary cortisol emerged as the most reliable and sensitive biomarker of acute psychosocial stress, consistently increasing during the Trier Social Stress Test (TSST) and decreasing during recovery. This finding highlights the central role of the hypothalamic–pituitary–adrenal (HPA) axis in stress regulation, since salivary cortisol emerged as the most sensitive endocrine marker of stress reactivity and recovery. Other HPA-related markers, such as ACTH and DHEA, showed less consistent results, mainly due to the limited number of available studies. In addition to the HPA axis, Man et al. (2023) found evidence for the involvement of the sympathetic–adreno-medullary (SAM) axis, with reliable increases in heart rate and salivary alpha-amylase during stress and decreases during recovery, indicating rapid autonomic activation in response to acute stress.

Beyond biological measures, the meta-analysis demonstrated emotional responses to the TSST. Particularly, a pronounced increase in negative affect during stress and a reduction during recovery (Man et al., 2023). However, findings in the emotional domain showed substantial variability

across studies, which was likely driven by differences in emotion measurement tools, sample characteristics, and limited statistical power. Positive effects showed only marginal and inconsistent changes (Man et al., 2023), suggesting that acute stress primarily amplifies negative emotional states rather than uniformly altering positive mood. Overall, these findings underline that acute psychosocial stress elicits coordinated but partially distinct responses across endocrine, autonomic, and emotional systems, with cortisol and negative affect representing the most reliable indicators of stress reactivity.

In addition, evidence suggests that positive affect can buffer physiological stress responses, whereas heightened negative affect is associated with stronger and more prolonged stress reactivity (Van Steenbergen et al., 2021). Thus, assessing affective changes using instruments as for example PANAS provides complementary information to endocrine measures, particularly when comparing task-specific stress effects.

Regarding the cognitive component, and according to Sandi (2013) acute stress has complex and sometimes opposing effects on cognitive functioning. According to integrative models of stress and cognition, the impact of stress depends on factors such as stress intensity, duration, controllability, and whether stress is intrinsic to the task or externally induced. Moderate acute stress can transiently facilitate simple or well-learned cognitive processes, whereas high stress levels tend to impair working memory, cognitive flexibility, and executive control functions that rely heavily on prefrontal cortex activity.

Within the TSST context, the mental arithmetic task imposes a substantial cognitive load under conditions of social evaluation and uncontrollability. This combination is likely to challenge working memory and attentional resources, potentially exacerbating stress-related cognitive impairments. In contrast, the speech task primarily involves self-presentation under social evaluation (Labuschagne et al., 2019), which may engage different cognitive and affective mechanisms.

Although the TSST is typically treated as a unitary stressor, its individual components may differ in their stress-inducing potential. Based on theoretical models of stress, cognition, and affect, it can be hypothesized that these task characteristics elicit differential stress responses. The following three hypotheses will guide the present study:

H1. Salivary cortisol responses will differ between the speech and mental arithmetic conditions.

H2. Autonomic stress responses (heart rate, RMSSD, salivary alpha-amylase) will differ between the speech and mental arithmetic conditions.

H3. Changes in affect (positive and negative affect) from pre- to post-TSST will differ between the speech and mental arithmetic conditions

- Luisa Taboada Albütz

Method

Participants

The total sample consisted of $N = 11$ participants (6 women, 5 men). Participants were quasi-randomized into two conditions based on their participant number (VPN): participants with odd numbers (VPN 1, 3, 5, etc.) received the speech condition, and participants with even numbers (VPN 2, 4, 6, etc.) received the math condition. Six participants were assigned to the speech task, and five participants to the math task.

Participants were recruited via convenience sampling using flyers in various buildings of the Friedrich-Alexander-Universität Erlangen-Nürnberg (FAU) in Erlangen, WhatsApp groups, email distribution lists, and personal contacts. The study was advertised as a study on presentation skills in English. A total of 91 individuals completed the screening questionnaire; however, this number also includes test trials conducted by members of the research team.

Demographic information was originally collected, including gender, biological sex, age, education, and current profession. However, due to unforeseen complications during data handling, not all screening data could be matched to the final participant sample. As a result, accurate demographic information beyond gender distribution (e.g., mean age or other descriptive statistics) cannot be reported.

Exclusion criteria were defined in advance and assessed during the screening process. Individuals were excluded if they did not provide informed consent, were younger than 18 or older than 40 years, or were not fluent in English (operationalized as not being able to hold a conversation in English). Further exclusion criteria included smoking more than five cigarettes per day in the past month or reporting regular drug use (e.g., marijuana, cocaine, opioids, LSD). Participants were also excluded if they had a body mass index (BMI) below 18 or above 30.

Additional exclusion criteria included the use of certain medications, such as medication for high blood pressure, heart failure, cardiac arrhythmia, or previous heart attacks, anticoagulant or anti-inflammatory medication, insulin, antidepressants, pain-relieving medication, cholesterol-lowering medication, hormone preparations, or medication intended to increase attention (psychostimulants). Furthermore, individuals were excluded if they reported medical conditions including diseases of the nervous system or eyes, acute symptoms of illness or inflammation, diseases

of the cardiovascular system, digestive system, liver, gall bladder, pancreas, skeleton, muscles, connective tissue, or skin, metabolic diseases, infectious or parasitic diseases, diseases of the kidneys, urinary tract, reproductive organs, or blood, as well as mental illness, cancer, or other serious medical conditions.

Participants were also excluded if they scored above 22 on the ADS-L questionnaire, indicating clinically relevant depressive symptoms. Prior participation in a stress test was initially defined as an exclusion criterion, but this requirement was later removed due to the low number of eligible participants. Participants were compensated either with €15 or one research participation hour (VP-hour).

- Judith Washof

Study Design

The study was conducted by students of the Friedrich-Alexander University Erlangen-Nuremberg as part of the "Digital Health Psychology" project seminar. We carried out a variety of measurements, which are described in detail in this paper. The data collection period was between November 26, 2025, and December 15, 2025. We conducted the tests between 1 p.m. and 6 p.m., with each test lasting approximately one hour. Our study was a laboratory study in which the participants' stress response was examined in the EmpkinS laboratory at Friedrich-Alexander University.

Our independent variable was the type of task in the TSST. We tested two conditions: the "math task" and the "speech task." The dependent variable was the participants' stress response. We measured the physiological stress response using an ECG, allowing us to record HR and HRV. In addition, we measured cortisol levels and alpha amylase via saliva samples and recorded body movements with a camera. We assessed the subjective stress response via self-report using the Positive and Negative Affect Schedule (PANAS; Watson et al., 1988).

We conducted a quasi-experimental study in which only participants who had successfully passed the prior screening were included. We employed a between-subjects design: participants were alternately assigned, based on the order of their appointments, to either the math task or the speech task. We conducted the study single-blinded, such that participants were not informed about the test condition, whereas both the experimenters and the committee were aware of it. Furthermore, we used a cross-sectional design, with a single data collection period obtained from the sample.

- Elina Lang & Christina Rogelj

Experimental Procedure

The participants were told to meet at the given address and time, where they were picked up by the investigator. The investigator welcomed the participant and led them to the waiting room, where they were asked to mute all technical devices. Then the participant was given a short and standardized introduction of the following procedure, in which the investigator explained the experiment's goal. It was told that the participant will take part in a stress test and that different kinds of physiological stress reactions (i.e. heart rate and cortisol) will be measured as well as information (i.e. positive and negative affect) by means of a questionnaire. Additionally, the investigator clarified that during the experiment it is not allowed to eat or drink except for some water right after providing a saliva sample.

After signing the consent form, the participant was put on the ECG-sensor. For a flawless measure of the heart rate, it was important to put the sensor appropriately onto the skin. After connecting the device with a corresponding application on the investigator's mobile phone, the participant was taught a signal (i.e. tapping twice on the sensor). Later in the analysis it should be possible to spot this signal, so it was used to mark the beginning of the stress test. This step was implied so that the participant can perform the signal right before starting the stress test, hence the exact starting point can be identified. In the next step, the participant fills out the first part of the PANAS (Watson et al., 1988). Afterwards they were instructed to take the first saliva sample (S0). Here, the participant saturated a cotton swab for 90 seconds. To provide an easy processing of the saliva samples, it was important not to touch the cotton swab or the edge of the tube with the lips as well as not chewing the swab. During the saliva sample, the investigator informed the committee via a Messenger that the test will start soon.

After conducting the first saliva sample, the standardized procedure of the TSST (Kirschbaum et al., 1993) was used. In the waiting room, the investigator asked about the participant's dream job. Afterwards, both the investigator and the participant went into the room next door, where the committee was waiting. This room was provided with the basic tools that are needed for the TSST. A marking was made on the ground right in front of the committee to mark the spot where the participant will stand during the test. After entering the room, the investigator handed over the mobile phone to the committee for later use. The participant was informed by the investigator to stand at the spot and redo the learned signal. The investigator explained in a standardized way the following procedure: The participant should imagine that they were invited to an interview for their dream job. For this interview the participant should convince the committee based on their personality traits. They would be provided with a preparation time of two minutes, during which they could make notes. These notes would later be not allowed to be used. The investigator also explained that speech and facial expression analysis will be performed with the video recording and that during the test, the

committee will observe and analyze non-verbal behavior. During the introduction to the task, the committee had enough time to adjust the camera on the mobile phone to the participant. After checking, if the participant understood the assignment correctly, the investigator left the room and the participant started their preparation. After two minutes, the active part of the committee interrupted the preparation and asked the participant to stand on the marking. After starting to record the video, either the speech or math task of the TSST would be performed.

In the speech condition, the participant was asked to start their presentation. First, the committee should let the participant speak freely as long as possible. In case that the participant finished their speech before the scheduled five minutes, the committee would remain silent for 20 seconds before pointing out that the participant still had some time. If this happened twice, the committee would begin to ask standardized questions. To induce stress, the committee was allowed to ask the participant to remain eye contact during their speech or to focus more on their personality traits instead of professional qualifications. In the math condition, the participant was asked to count backwards from 2043 in steps of 17. This should happen as quickly and accurately as possible. If the participant made any mistake, the committee would point it out and the participant had to start again. To induce stress, the participant should pronounce the numbers in full. In both conditions, after the time is over, the committee thanked the participant, stopped the video recording and asked the participant to leave the room, where the investigator would pick them up. It is important to note that during the TSST only the active member of the committee would interact with the participant. Both the active and passive members were wearing white lab coats and were allowed and asked to take notes during the TSST to induce more stress.

Back in the waiting room, the second saliva sample (S1) was conducted one minute after the TSST. In the next step, the second part of the PANAS was filled out. Five minutes after the TSST, the recording of the ECG-sensor was stopped and the participant took off the sensor. Ten minutes after the TSST, the third saliva sample (S2) was conducted. Twenty minutes after the TSST, the fourth and last saliva sample (S3) was executed. In the meantime, the way of compensation could be discussed and accordingly handled. After the last saliva sample, the participant was debriefed and was able to meet the committee.

- Florian Hofer

Measures

Stress responses were examined using both self-report and physiological measures. These measures are described in the following section.

- Sophie Stadler

Self-Report Measure

Subjective affective stress responses were measured using the Positive and Negative Affect Schedule (PANAS; Watson et al., 1988). The PANAS assesses two independent dimensions of affective states: positive affect (PA) and negative affect (NA). It comprises 20 items, 10 reflecting positive affect and 10 reflecting negative affect. Example items for positive affect include active, interested and enthusiastic, whereas example items for negative affect include anxious, irritable, and ashamed. Participants rated each item on a five-point Likert scale, indicating the extent to which they currently experienced the respective affect, ranging from 1 (very slightly or not at all) to 5 (extremely).

Within the present study, the PANAS was administered twice, once before the TSST and once after the TSST, in order to capture changes in affect. Separate scores for positive and negative affect were calculated by aggregating the respective items (Watson et al., 1988). Changes in negative affect are considered a sensitive indicator of emotional stress responses during the TSST (Man et al., 2023).

- Nora Straschewski

Physiological Measures

Physiological stress responses were indexed by salivary biomarkers and measures of cardiovascular activity, which are described in detail below.

- Sophie Stadler

Cortisol and Salivary Alpha-Amylase.

To assess activity of the HPA axis, cortisol was measured (Smyth & Clow, 2020). Salivary alpha-amylase (sAA) was used as a biomarker of sympathetic nervous system activity (Nater & Rohleder, 2009). Saliva samples were collected using Salivettes (Sarstedt, Nümbrecht, Germany). A total of four saliva samples were obtained from each participant. Samples were stored in a refrigerator at -18 °C and transported to the laboratory for analysis after completion of data collection. There, the samples were centrifuged at $2000 \times g$ and 20 °C for 10 minutes. A chemiluminescence immunoassay (CLIA; IBL, Hamburg, Germany) was used to measure cortisol concentrations, while sAA was analyzed using an in-house enzyme kinetic assay (DiaSys Diagnostic Systems GmbH, Holzheim, Germany). Both assays followed procedures described previously (e.g., Hauck et al., 2022). Cortisol concentrations are reported in nmol/L and sAA activity in U/mL.

- Sophie Stadler

Heart Rate and Heart Rate Variability.

Heart rate (HR) and heart rate variability (HRV) were recorded as physiological indicators of stress responses. HR refers to the number of heartbeats per minute (beats per minute, bpm), whereas HRV reflects beat-to-beat fluctuations in interbeat intervals. Changes in HR and HRV are considered indicators of autonomic stress responses, as stress is associated with increased sympathetic nervous system activity and reduced parasympathetic activity (Kim et al., 2018). Cardiac activity was recorded using an electrocardiogram (ECG) with a wearable ECG system (NilsPod). The recording was

obtained via an ECG chest belt worn underneath the participant's clothing at heart level, with the NilsPod sensor positioned approximately over the sternum.

From the recorded ECG data, several HRV parameters were calculated, including the standard deviation of Normal-to-Normal intervals (SDNN) and the root mean square of successive differences (RMSSD). SDNN reflects overall heart rate variability, whereas RMSSD primarily represents parasympathetic activity (Task Force of the European Society of Cardiology and the North American Society of Pacing and Electrophysiology, 1996). Additionally, mean heart rate and maximum HR increase during the stress task were derived from the ECG data as indicators of physiological stress reactivity.

- Nora Straschewski

Statistical Procedure

PANAS Questionnaire

The affective responses were assessed using the PANAS (Positive and Negative Affect Schedule) questionnaire. Both positive affect (PA) and negative affect (NA) scores were analyzed as dependent variables. To account for the repeated-measures structure of the data, a linear mixed-effects model was employed. Time was treated as a within-subject factor, while condition was included as a between-subject factor. The interaction between time and condition was also modeled to examine differential changes across conditions over time. Random intercepts were included for each participant to account for inter-individual variability in baseline affect levels. This approach allowed for robust estimation of fixed effects while controlling for within-subject correlations. All statistical analyses were performed separately for positive and negative affect scores.

Salivary cortisol and alpha amylase

Salivary biomarkers were analyzed to assess physiological stress responses. Cortisol concentration (nmol/L) and salivary α -amylase activity (U/mL) were treated as dependent variables. For both biomarkers, a 2×2 between-subjects analysis of variance (ANOVA) was conducted and experimental conditions as independent factors. Separate ANOVAs were performed for each parameter to assess differential physiological responses.

Electrocardiogram

A one-way between-subjects analysis of variance (ANOVA) was conducted with experimental condition as the independent factor. This analysis was used to examine differences in cardiovascular and heart rate variability (HRV) parameters between conditions. Maximum HR increase and mean HR were evaluated as indicators of cardiovascular activation, while SDNN and RMSSD were analyzed as time-domain measures of heart rate variability, reflecting autonomic nervous system regulation. Separate ANOVAs were performed for each parameter to assess condition-specific effects.

- Vindu R

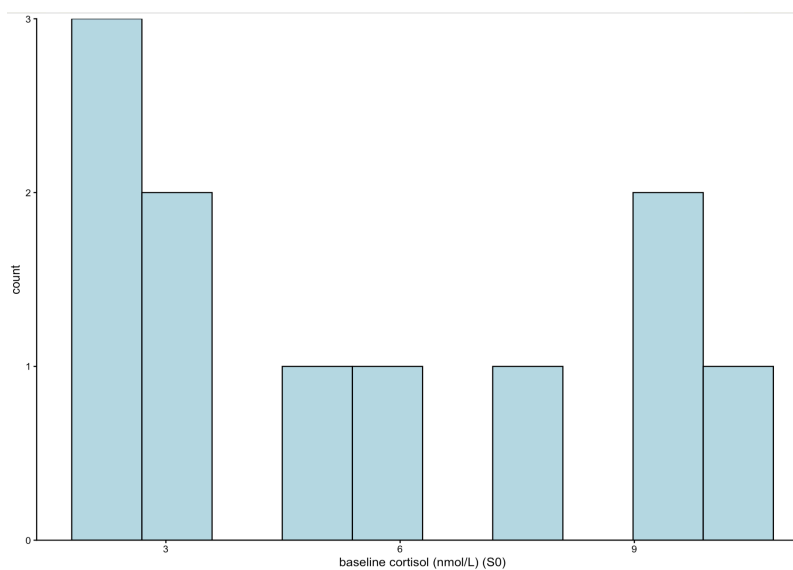
Statistical Analysis

Cortisol

We checked cortisol for outliers and normality of distribution. First, we analyzed baseline cortisol levels (S0) for outliers by calculating z-scores, considering z-scores above +3.29 or below -3.29 as outlier cases. There were no significant outliers at baseline (z-scores ranged from -1.04 to 1.50). Similarly, visual inspection of the frequency distribution histogram did not indicate the presence of outliers (see Figure 1).

Figure 1

Frequency distribution of baseline cortisol values (S0)



Next, we performed outlier analyses for each sampling time point separately using z-scores. None of the z-scores exceeded the cut-off for significant outliers at any time point. Normality of distribution was assessed by visual inspection of histograms and computation of the Shapiro-Wilk test for each sampling time point separately. The assumption of normality of distribution was violated for S1, S2 and S3, indicated by significant Shapiro-Wilk tests. Thus, we log-transformed cortisol for each time point separately. After log-transformation, Shapiro-Wilk tests per variable ceased to be significant.

As a response index for cortisol, we calculated maximum increase by subtracting log-transformed baseline cortisol levels (S0) from the highest value observed at either S1 (+1 minute after TSST), S2 (+10 minutes after TSST) or S3 (+20 minutes after TSST). To test our main hypothesis, we computed a 2×2 between-subjects ANOVA with the between-subjects factors condition (speech vs. math) and sex (male vs. female) and cortisol maximum increase as dependent variables. Statistical assumptions for a between-subjects ANOVA were tested. Homogeneity of variances for maximum increase was tested across the four conditions by computing a Levene's test.

There was no evidence of a violation of homogeneity of variances, $F(3, 7) = 1.01, p = .44$. Due to the very small number of observations per condition, we did not test for normality of cortisol maximum increase per condition.

- Louisa Quost

Salivary Alpha Amylase

Data preprocessing and analysis was performed by R in RStudio. Graphs were created with ggplot. For data preprocessing, we extracted the largest difference between sAA measurement at t0 = baseline in comparison to sAA measurements at the remaining timepoints t1 to t3. In other words, we aggregated the measurements of sAA as maximum increase (sAAmax) to enable statistical analyses in accordance with the small sample size. Values beyond the cutoff of 2.5 standard deviations were excluded as outliers. Normality of distribution and homogeneity of variance was checked by visual inspections of residuals in a histogram and Q-Q plot. We performed a Shapiro-Wilk normality test and a Levene's test. We computed a two-way ANOVA with condition (math vs. speech) and sex (male vs. female) as independent variables and alpha amylase as dependent outcome variable. Statistical significance was set at $\alpha = 0.05$.

- Anna Yi

Electrocardiogram (ECG)

ECG signals were analyzed to derive heart rate (HR) and heart rate variability (HRV) metrics for each participant. HRV was quantified using standard time-domain measures, including the standard deviation of normal-to-normal intervals (SDNN) and the root mean square of successive differences (RMSSD), which reflect overall variability and short-term parasympathetic modulation of cardiac activity, respectively (Task Force of the European Society of Cardiology, 1996; Shaffer & Ginsberg, 2017). For each participant, mean heart rate (mean HR, bpm), maximum increase in heart rate relative to baseline (max increase HR, bpm), SDNN (ms) and RMSSD (ms) were computed as outcome measures. Mean HR was compared descriptively between conditions, while inferential statistics were applied to the primary physiological outcomes. A one-way between-subjects analysis of variance (ANOVA) was conducted separately for each outcome variable to test for differences between the two experimental conditions (speech vs. math). Condition was treated as a between-subjects independent variable, and the dependent variables were maximum increase HR (heart rate reactivity), RMSSD (primary HRV measure) and SDNN (secondary HRV measure). Max increase HR and RMSSD were defined as primary outcomes, as they reflect acute stress reactivity and parasympathetic regulation, respectively, whereas SDNN was analyzed as a secondary indicator of overall heart rate variability (Berntson et al., 1997; Shaffer & Ginsberg, 2017).

The Trier Social Stress Test (TSST) paradigm is known to elicit autonomic and endocrine stress responses, including increases in heart rate and reductions in HRV, particularly under

social-evaluative threat and cognitive load (Kirschbaum et al., 1993; Dickerson & Kemeny, 2004). Statistical significance was evaluated at $\alpha = 0.05$. All analyses were performed using Python.

- Meghana

Physiological Data Processing

Physiological data were recorded using a NilsPod wearable sensor system, which collects electrocardiographic (ECG) signals as well as inertial measurement unit (IMU) data, including acceleration and gyroscope signals. The raw data were exported as binary (.bin) files and processed in Python using the BioPsyKit toolbox (Richer et al., 2021). Recordings were loaded based on an Excel spreadsheet containing participant metadata (e.g., participant ID and experimental condition) along with the corresponding file names.

ECG data was processed using the EcgProcessor module from BioPsyKit. When necessary, the ECG signal was inverted to ensure that R-peaks were positive. Standard preprocessing steps were applied, including filtering and R-peak detection, which were performed simultaneously. To avoid instability at the beginning of recordings, the first 4 s of each signal were discarded. Baseline drift was removed using a rolling median filter with a 1.5-s window, and the detrended signal was normalized using the 99th percentile of the signal amplitude to account for inter-individual variability.

IMU signals were used to identify motion artifacts. To reduce high-frequency noise while preserving physiologically relevant movement, acceleration and gyroscope data were filtered using a fourth-order low-pass Butterworth filter with a cutoff frequency of 20 Hz. The magnitude of the filtered acceleration and gyroscope signals was computed and smoothed using a 1-s rolling mean window. Motion periods were then identified based on the rolling standard deviation calculated over a 0.5-s window. A threshold was defined as the mean plus two standard deviations, and periods exceeding this threshold were classified as high-motion segments. R-peaks occurring during these segments were excluded to improve the accuracy of heart rate (HR) and heart rate variability (HRV) estimates.

Heart rate was calculated from detected R-peaks and expressed in beats per minute (bpm). Heart rate variability measures were derived from RR intervals (i.e., the time between successive R-peaks). Two commonly used time-domain HRV metrics were computed: the standard deviation of normal-to-normal intervals (SDNN) and the root mean square of successive differences (RMSSD) (Shaffer & Ginsberg, 2017). In addition, an HRV-based stress score was calculated from RMSSD values, ranging from 10 ms to 100 ms, with lower HRV values indicating higher physiological stress.

These preprocessing steps resulted in clean ECG signals, motion-corrected R-peak detection, and reliable HR and HRV measures for subsequent statistical analyses across participants and experimental conditions.

- Amalna Jose

PANAS

Data were analyzed using *Python* (packages: *pandas*, *scipy*, and *statsmodels*). Positive Affect (PA) and Negative Affect (NA) scores were computed according to standard PANAS scoring procedures (Watson et al., 1988).

Responses were first screened for validity, and values outside the scale range (1–5) were treated as missing. For each subscale, participants with ≤ 2 missing items had missing values replaced using person-mean imputation, while subscales with ≥ 3 missing items were treated as missing.

PA and NA scores were calculated separately for pre- and post-measurements by summing the respective items. Change scores (Δ PA, Δ NA) were computed as the difference between post- and pre-scores. Descriptive statistics (means and standard deviations) were computed for all variables by condition and time point. To assess baseline equivalence between groups (*math* vs. *speech*), independent-samples t-tests (Welch correction) were conducted for PA and NA at pre-measurement. To examine changes over time, linear mixed-effects models were fitted separately for PA and NA, with Time (Pre vs. Post), Group (*math* vs. *speech*), and their interaction as fixed effects, and participant ID as a random intercept (Bates et al., 2015). Effect sizes for group differences in change scores were calculated using Cohen's *d*. The significance level was set at $\alpha = .05$.

- Kimia Sedighi

Results

Table 1

Descriptive statistics

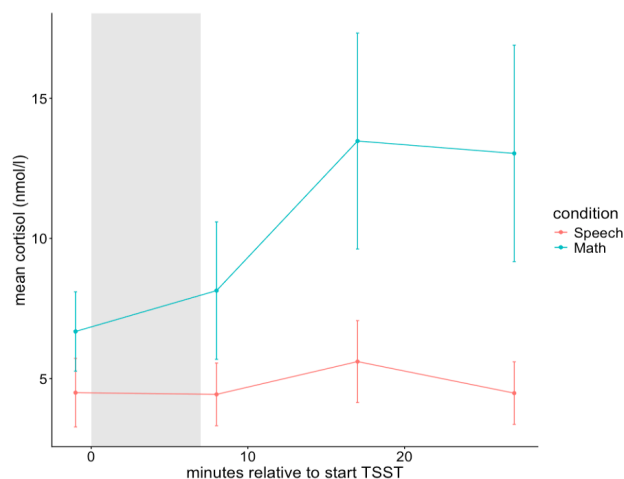
condition	n	cortisol max. increase (nmol/l); M(SD)	sAA max (U/ml); M(SD)	PA pre; M(SD)	PA post; M(SD)	NA pre; M(SD)	NA post; M(SD)	HR mean (bpm) M(SD)
speech	6	0.25 (0.16)	40.2 (60.3)	31.17 (6.97)	29.33 (11.41)	12.50 (2.35)	14.83 (6.62)	95.91 (14.10)
math	5	0.61 (0.40)	40.2 (60.3)	31.60 (9.18)	31.20 (9.15)	14.20 (3.96)	16.20 (6.26)	78.61 (10.54)

Cortisol

There was no significant difference between the math and speech condition in terms of baseline cortisol levels, $t(9) = -1.17$, $p = 0.27$. To test whether the TSST elicited a significant cortisol response, a univariate repeated-measures ANOVA was conducted, indicating a significant change in cortisol levels over time, $F(1.19, 11.9) = 5.35$, $p < .05$. The hypothesis that maximum cortisol increase would significantly differ between the math and the speech task was not supported. There was no significant main effect of condition on maximum cortisol increase, $F(1, 8) = 4.20$, $p = .08$, nor was there a significant main effect of sex, $F(1, 8) = 0.53$, $p = .49$. However, there was a tendency for higher cortisol responses in the math condition compared to the speech condition (see Figures 2, 3, 4).

Figure 2

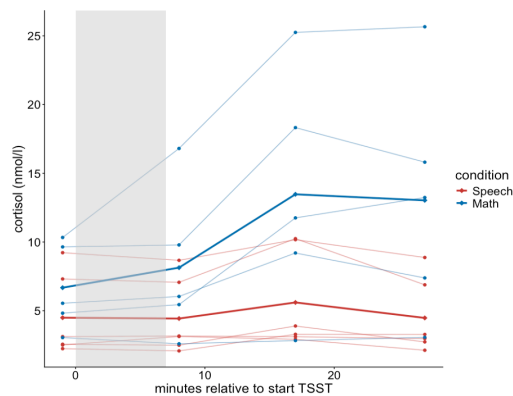
Mean cortisol levels over time by condition



Note. This graph depicts untransformed mean cortisol levels and the standard error of the mean at baseline and 1, 10 and 20 minutes relative to the end of the TSST for the speech and math condition separately. The gray bar indicates when the TSST took place.

Figure 3

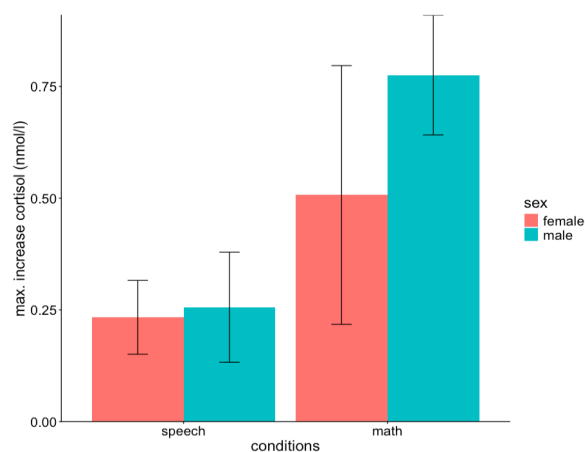
Individual cortisol trajectories



Note. This graph depicts untransformed mean cortisol levels at baseline and 1, 10 and 20 minutes relative to the end of the TSST for each participant separately. Bold lines indicating condition means.

Figure 4

Maximum cortisol increase per condition and sex



Note. This graph depicts mean maximum cortisol increase values and the respective standard error of the mean for each condition divided by sex.

- Louisa Quost

Salivary Alpha Amylase

There were no outliers beyond the cutoff of 2.5 standard deviations. We further checked the statistical assumptions for two-way factorial ANOVA by visual inspections of residuals in a histogram and Q-Q plot. Residuals appeared normally distributed and the Q-Q plot suggested normality as well.

The Shapiro-Wilk normality test returned false at a significance value of $p = .42$. Therefore we can assume normal distribution of sAAmax. Homogeneity of variance was successfully supported by a nonsignificant Levene's test ($p = .7$; F -value = .49). We can assume that the prerequisites for two-way ANOVA are fulfilled.

Mean sAA appeared similar across math and speech conditions, with larger spread in the math condition. When comparing mean sAA between timepoints t0 through t3, means appeared to show a slight increase from t0 to t2, followed by a decrease at t3 (Fig. 5). We did not find support for the hypothesis that maximum increase of alpha amylase differs between math and speech conditions. There was neither a main effect for condition math vs. speech ($F(1,8) = .16$, $p = .68$) nor sex ($F(1,8) = 1.04$, $p = .34$). Despite the difference in mean sAAmax between female and male participants in the speech condition upon visual inspection (see fig. 6 & 7), the interaction term turned out insignificant as well ($F(1,8) = 1.63$, $p = .23$).

Figure 5

Comparison of mean alpha amylase across sample timepoints

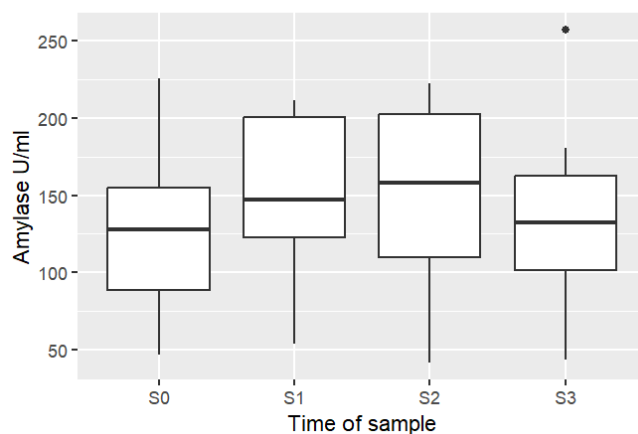


Figure 6

Mean sAA (i.e. non-aggregated data) across timepoints, comparing math and speech condition

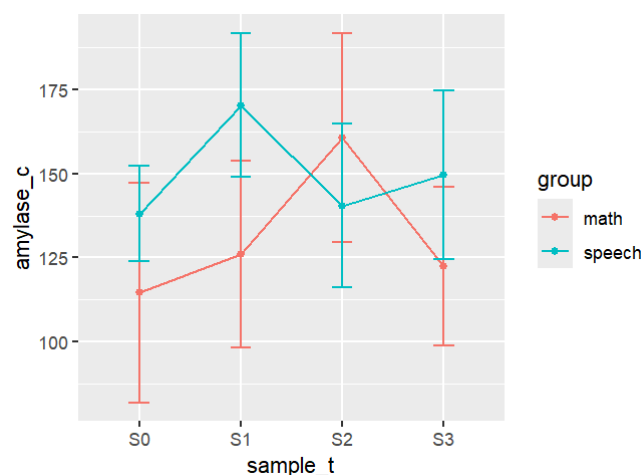
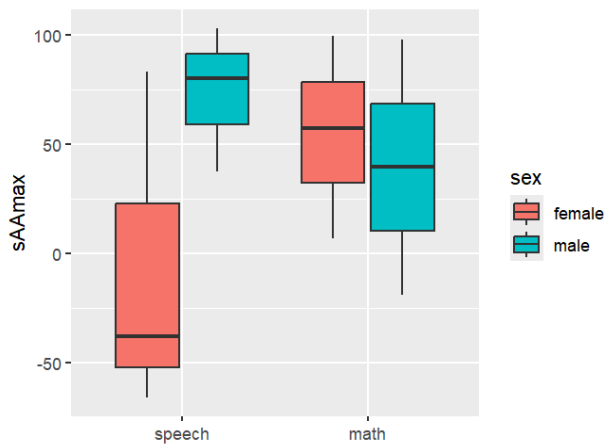


Figure 7*Mean sAAmax per condition and sex*

- Anna Yi

Heart Rate (HR) and Heart Rate Variability (HRV)

HRV is not a single value; it is quantified using metrics such as SDNN and RMSSD, which capture overall and short-term variability of RR intervals. Table 1 has mean HR values.

ANOVA

For maximum increase in HR, the ANOVA showed no significant effect of condition, $F(1, 9) = 0.64$, $p = .44$, indicating that peak heart rate reactivity did not differ reliably between the speech and math conditions. Although the TSST typically induces strong cardiovascular responses, variability between individuals can be substantial, particularly in small samples (Kirschbaum et al., 1993; Dickerson & Kemeny, 2004).

For SDNN, the ANOVA also revealed no significant difference between conditions, $F(1, 9) = 0.14$, $p = .72$, indicating that overall heart rate variability was comparable across the two tasks. Similarly, RMSSD did not differ significantly between conditions, $F(1, 9) = 0.27$, $p = .62$.

Taken together, the results indicate that although the tasks elicited physiological activation, the difference between speech and math conditions was not statistically significant for heart rate reactivity or HRV measures. This pattern suggests that individual differences in stress responsiveness may have been larger than the effect of task condition, which is commonly observed in psychophysiological stress studies with limited sample sizes (Dickerson & Kemeny, 2004; Laborde et al., 2017).

- Meghana

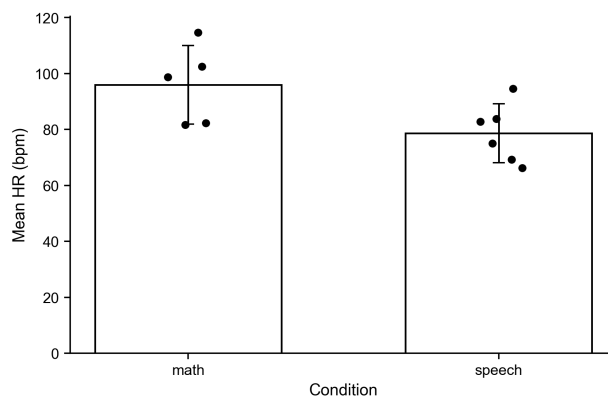
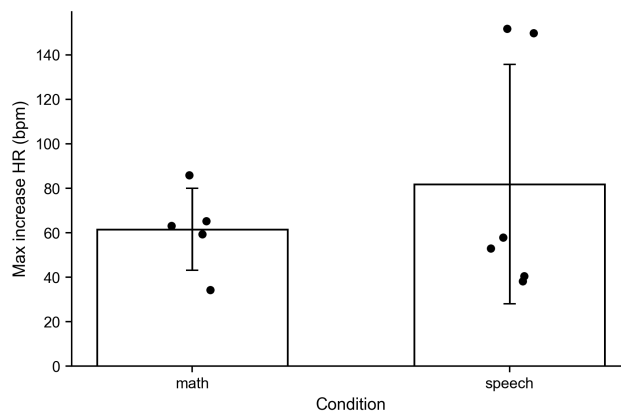
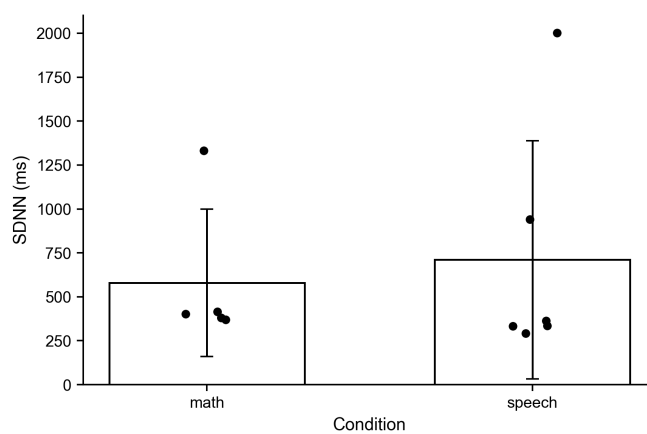
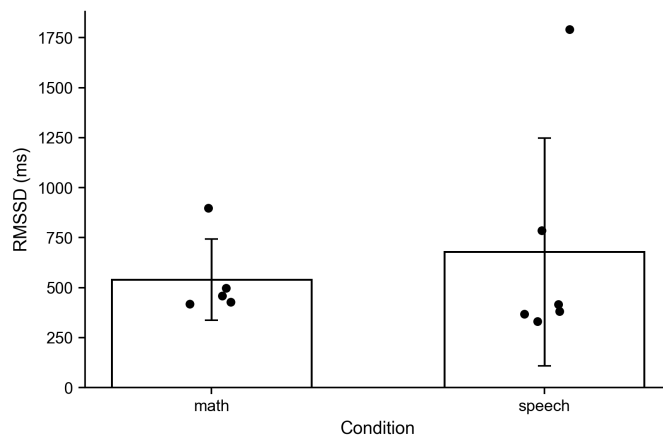
Figure 8*Mean HR by condition (speech vs math)***Figure 9***Max increase HR by condition (speech vs math)***Figure 10***SDNN by condition (speech vs math)*

Figure 11*RMSSD by condition (speech vs math)*

- Amalna Jose

Inference

Table 2

Outcome measure	F(1,9)	p-value	Statistical Inference	Physiological Interpretation
Max increase HR (bpm)	0.64	0.44	Not significant	Peak heart rate reactivity does not differ reliably between speech and math; large inter-individual variability dominates.
SDNN (ms)	0.14	0.72	Not significant	Overall, heart rate variability is comparable between conditions.
RMSSD (ms)	0.27	0.62	Not significant	Short-term parasympathetic modulation does not differ significantly between speech and math.

- Meghana, Amalna Jose

PANAS

Descriptive statistics for Positive Affect (PA) and Negative Affect (NA) are presented in Table 1.

Baseline Equivalence.

There were no significant differences between the math and speech groups at baseline.

- Positive Affect: $t(7.40) = 0.09, p = .933$
- Negative Affect: $t(6.25) = 0.84, p = .430$

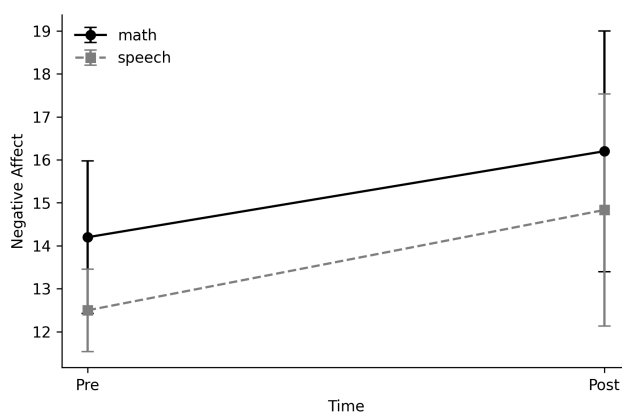
This indicates that the groups were comparable before the manipulation.

Negative Affect.

The linear mixed-effects model revealed no significant main effect of Time ($p = .420$), no significant main effect of Group ($p = .657$), and no significant Time $\times$ Group interaction ($p = .921$). This suggests that negative affect did not change significantly over time and was not influenced by group. The effect size for change in negative affect was negligible (Cohen's $d = -0.06$). As illustrated in Figure 10, negative affect scores remained relatively stable across time in both conditions.

Figure 12

Negative affect pre-post by condition. Error bars represent ± 1 SEM

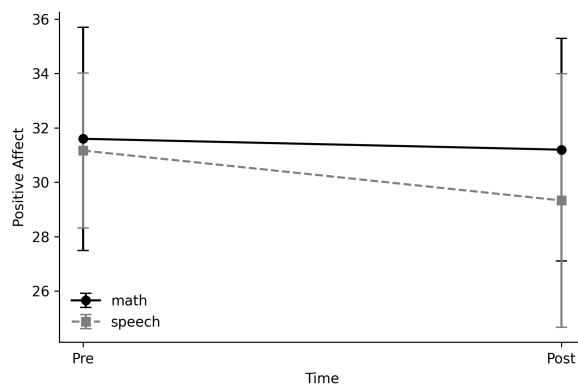


Positive Affect.

The linear mixed-effects model showed no significant main effect of Time ($p = .813$), no significant main effect of Group ($p = .741$), and no significant Time $\times$ Group interaction ($p = .530$). Thus, positive affect did not significantly differ across time or between groups. The effect size for change in positive affect was small-to-moderate (Cohen's $d = 0.38$). As shown in Figure 11, positive affect scores exhibited only minor changes across time in both groups.

Figure 11

Positive affect pre-post by condition. Error bars represent ± 1 SEM



Overall, there were no statistically significant changes in positive or negative affect, no differences between conditions (math vs. speech), and no significant interaction effects.

- Kimia Sedighi

Discussion

Specific Findings

The statistical analyses of salivary cortisol, salivary alpha amylase, HR, HRV and positive and negative affect did not support a different stress response to the math or speech condition of TSST.

Yet, there seemed to be a slight divergence among the stress systems regarding differential intervention effects. All autonomic nervous system markers converged in that the interventions did not differentially impact stress reactivity, as indicated by non-significant differences in sAA levels, HR and HRV reactivity. Even though the difference in cortisol reactivity between conditions did not reach statistical significance, a tendency was observed for maximum cortisol increase to be higher in the math compared to the speech condition. Given the very small sample size, there might have been insufficient power to detect an existing effect. In the same way, the small sample size increases the risk of group differences created by chance. In both respects, to draw any conclusions, a study with greater statistical power is needed. Yet, this preliminary trend observed for cortisol is in line with earlier findings suggesting that HPA-axis reactivity is further exacerbated by adding a challenging cognitive task to the social evaluative component (Dickerson & Kemeny, 2004).

Consistent with these physiological findings, self-reported affect measured via PANAS also did not differ significantly between conditions or across time. Neither positive nor negative affect showed significant changes from pre- to post-measurement, and no interaction effects between time and group were observed. This convergence between subjective and physiological measures

strengthens the interpretation that both stress conditions elicited comparable overall stress responses.

Although no statistically significant effects were found for PANAS, the observed small-to-moderate effect size for changes in positive affect may indicate a subtle trend that did not reach significance, potentially due to the limited sample size. This suggests that minor psychological differences between stress conditions cannot be entirely ruled out.

Mean sAAMax in response to the speech-based intervention appeared to differ between female and male participants. If confirmed as a significant effect, this pattern might suggest a gender-moderated response to social stress.

SDNN reflects global autonomic regulation, which may not always show condition-specific effects in short experimental sessions (Task Force, 1996; Shaffer & Ginsberg, 2017). As SDNN and RMSSD did not differ significantly between conditions suggesting that short-term parasympathetic modulation was not reliably affected by the experimental manipulation. RMSSD is particularly sensitive to vagal activity and may show high inter-individual variability in stress paradigms (Berntson et al., 1997; Laborde et al., 2017).

- Anna Yi, Kimia Sedighi, Louisa Quost, Meghana

Limitations

Due to its design, the study has several limitations that may restrict the interpretability and validity of the results. First, there are limitations regarding the sample. Although the gender distribution was roughly balanced, the overall sample size of six women and 5 men was very small. Such a small sample size reduces statistical power and increases the likelihood of committing a type-2-error, which could lead to incorrect conclusions because the study may fail to detect an effect that truly exists (Lakens, 2022). Although such errors can never be entirely avoided, their likelihood can be reduced by increasing the sample size, since larger samples are less likely to differ substantially from the population (Banerjee et al., 2009). Furthermore, the representativeness of the sample was limited, which threatens external validity (Shadish et al., 2002). Given that the participants were drawn from the immediate social environment of the experimenters, it can be assumed that they also shared similar educational backgrounds. The educational backgrounds may have influenced the stress response following the TSST, as previous research suggests that individuals with a higher level of education tend to experience lower stress in performance-related tasks compared to those with lower levels of education (Fiocco et al., 2007). Additionally, the participants were recruited by the seminar participants themselves, who also served as the experimenters in this study. As a result, the sample primarily consisted of friends and family members. The magnitude of the stress response elicited by the TSST depends on the level of social-evaluative threat experienced by the participants. When this perceived threat is lower, for example due to a less intimidating social context, the stress response may be attenuated (Frisch et al., 2015). Such a reduction may have occurred in this study because of the familiarity between the participants and the experimenters.

In addition to sample-related issues, the study also shows some methodological limitations that may have influenced the stress response independently of the dependent variable, i.e., regardless of the way the TSST was administered. The quasi-experimental design and lack of randomization increase the possibility that confounding variables affected the results, but those were neither identified nor statistically controlled for. Individual differences such as the presence and severity of social anxiety could have affected the stress responses of the participants after the TSST. Previous research pointed out that socially anxious people tend to feel pressured to create a positive appearance and the mere experience of being observed can trigger a stress response (Schlenker & Leary, 19829). Other personality differences can protect an individual against stress, for example a high core self-evaluation, a construct that includes personality traits like global self-esteem and emotional stability (Schneider et al., 2025). If such individual differences are unevenly distributed across groups, they may contribute to systematic group differences, which could influence the results.

The operationalization also comes with certain limitations. For example, the study was conducted entirely in English, which was a foreign language for most participants. Performing a

stressful task in a non-native language, such as during the TSST, can itself increase perceived stress levels, because the situation might appear more stressful and threatening for the participants (Hauck et al., 2022). Additionally, the time of the day at which the study was conducted may have influenced the results, as physiological stress responses tend to fluctuate over the course of the day (Yamanaka et al., 2019). Both hypothalamic-pituitary-adrenal (HPA) axis activity, measured via cortisol, and sympathetic nervous system (SNS) activation, measured via salivary alpha-amylase (sAA), tend to be higher in the morning (Yamanaka et al., 2019; Schneider et al., 2025). The use of sAA is associated with limitations, given its high intraindividual variability and its sensitivity to external factors, including the time of the day and food intake (Nater et al., 2007). Additionally, it is not the absolute value but only increases in sAA levels that correlate with psychosocial stress. This may be due to already existing interindividual differences in baseline sAA levels (Rohleder et al., 2004). Similarly, cortisol levels also vary naturally, with higher baseline values in the morning (Yamanaka et al. 2019). Taken together, these circadian influences and interindividual differences highlight that neither of those biomarkers used in this study provides a perfectly stable measure of stress. Additionally, the results of the ECG conducted as part of the study may also have been affected by the participants' fitness levels, as higher physical fitness appears to be associated with smaller changes in heart rate following the TSST (Mücke et al., 2018). Moreover, self-report measures, such as the Positive and Negative Affect Schedule (PANAS; Watson et al., 1988) that we used in this study, are prone to various biases like response tendencies. Because of that, systematic differences in response patterns may have occurred and went unnoticed. In addition, experimenter effects may have influenced the results, although they were not explicitly controlled for. The TSST requires a high degree of standardization (Narvaez Linares et al., 2020), whereas the experimenters had little to no experience with it prior to this study. The lack of experience may have resulted in inconsistent instructions and varying degrees of strictness within the TSST jury, potentially leading to differences in the intensity of the induced stress response.

In summary, the combination of a small, non-representative sample, a quasi-experimental design with no randomization, potential confounding variables, limitations regarding the operationalization, unaccounted experimenter effects and the increased likelihood of missing real effects due to limited statistical power all limit the reliability and generalizability of the findings. While the results provide valuable information on psychosocial stress, these limitations should be considered when interpreting the data.

- Marina Cörlin

Implications and Conclusion

The analysed data doesn't allow for a conclusion that social- or respectively more cognitive focused stress (in the math condition) result in a different stress reaction. Or more concrete: Solving arithmetic problems in front of experimenters doesn't induce a different stress response than talking about yourself. But as mentioned earlier, there were signs that the cortisol was higher after the math condition, also regardless of the gender, which could be a first hint, that the different tasks do in fact illicit a different response in the stress system; Of course the statistical analysis did not reveal a significant effect, so we can't make clear conclusions and this guess is only exploratory as a signpost for future studies. In such, possibly real effects could be detected with a sufficient sample size. Having only five respectively six participants per group in the statistical analysis of the present study makes it really hard to find an effect that is actually there. Given the example of an independent t-test, the likelihood of spotting an actual effect of $d = 0.5$ more than doubles if you increase the group size from 5 to 15 and increases further if you test more participants (Lakens, 2022). Of course just blindly adding more participants to increase the statistical power would lead to an increased beta-error, but aiming for a sufficient sample size in future studies is recommended

After taking into account the importance of sample size, it would be interesting to look at possibly confounding variables: Some candidates that could have an impact are trait neuroticism or also (subclinical) social anxiety. For example it is possible that socially anxious people have a very detectable stress response just from standing in front of observers alone (Schlenker & Leary, 1982), which could influence the effect of the condition (math vs. speech) heavily. If you look closer into theories about social anxiety (e.g. Bögels & Mansell, 2004), social anxiety could even have a quite significant impact in combination with the given task: For example, solving mathematical problems could shift the participants' attention away from the self-presentation that is characteristic for the condition (Clark & Wells, 1995) and actually lower rather than increase the effect of the perceived social stressor, since a lot of the attention is consumed by the challenging math task. For example, in recent studies it's been shown that shifting away your attention from yourself can lower anxiety in presentations (Wechsler et al., 2021). In future studies, intra-individual variables that could be confounded with the stress reaction should be taken into account, so it can be examined correctly, how the math and speech task elicit stress. To further examine how the stressors are perceived, it would also be useful to ask the participants about *what exactly* caused stress for them; this could help to understand different kinds of stressors that can be used in future stress-tests better. Gaining more insight about the association of individual aspects (e.g. level of social anxiety/shyness) and experiences about what exactly causes stress in stress tasks (e.g. self-presentation or failing at math tasks etc.) could lead to a better possibility of controlling for the effect of individual characteristics.

Another interesting aspect about our study are the results regarding biological sex: In previous studies, women were found to sometimes show a smaller cortisol-reaction to TSST in comparison to men albeit the subjective stress did not differ (Reschke-Hernández et al., 2017). Such findings are in contrast to our study, where men and women showed no difference regarding the task type in terms of the physical stress response as well as the subjective (PANAS). Given the small sample size, no clear conclusions can be made about this. Future studies should still look at sex-differences in the relationship of perceived and physiological stress and how they are linked to different task types to bring more light into previous findings. Getting back to our results, it would be interesting to compare characteristics of our sample with the sample of the mentioned study of Reschke-Hernández et al. (2017). Maybe differences in perceived self-confidence about mastering different kinds of tasks could play a role: In the scientific community gender and math-anxiety has been evaluated a lot, although the differences seem to be smaller than often expected (Barroso et al., 2021). Still, self concept about arithmetic abilities and/or the fear of being evaluated about specific tasks could have played a role in our study. In the future, such specific fears should be evaluated too, which could isolate the conditions' effect better.

Another improvement for future studies that was already mentioned in the study's limitations would be to recruit a more diverse sample. Not only would it increase the external representativeness (Shadish & Campbell, 2002) but also result in a more diverse educated background overall, which could have an influence on how the situation of speaking in front of strange people is perceived.

To conclude, our study does not support the claim that holding a speech task or solving math in front of other people cause different stress responses. However these findings should be looked at with caution because of the study's limitations regarding the homogeneous and small sample size and the lack of control of confounding variables. Future studies should repeat the procedure with a more diverse sample and take into account neuroticism, social anxiety and physical fitness as well as the daytime of the testing to be able to isolate the effect of the condition better. Furthermore exploring sex-differences regarding the relationship between physical stress reaction and perceived stress could increase the understanding of how different tasks induce stress even better.

- David Blatterspiel

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